

Matthew Zhou

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Education

McGill University

Bachelor of Engineering in Software Engineering Co-op

Aug 2025 – Expected 2029

GPA: 3.9/4.0

Skills

Languages: Python, C++, Java, Javascript, HTML/CSS/SCSS

Libraries & Tools: Git, React, Docker, Three, Leaflet, Github, Node, PostgreSQL, React Flow, LaTeX

Experience

Ciena

Ottawa, Ontario

Reliability Engineer Intern | Python, VBA, ChromaDB, RAG

May 2026 – Aug 2026

- Built a FIT data ETL pipeline normalizing **5** card-type workbooks into a **ChromaDB** vector database, enabling semantic similarity matching across **500+** empirical test records.
- Extended a **hybrid-RAG** FMECA prepopulation system with a parallel FIT vector database layer, surfacing empirical ground-truth results alongside AI-generated field predictions.
- Automated SIR coupon resistance measurements interfacing Python with a **Hioki** IR meter and **41-channel** relay controller with persistent session-aware logging.
- Performed capacitor testing and PCB resoldering in a hardware lab, validating electrical components for circuit board reliability engineering.

McGill Formula SAE Electric

Montreal, Quebec

Driverless/Autonomous Member | C++, ROS 2, Python, Docker

Sept 2025 – Present

- Training **YOLOv8**-based deep learning model for live cone detection and classification under different lighting conditions.
- Developing **ROS2** nodes to process sensor data and manage high-frequency changes for real-time autonomous navigation.

VEX Team 50226A

Ottawa, Ontario

Software Team Lead / Mechanical Member | C++, CAD

Oct 2023 – Feb 2025

- Developed autonomous and teleoperated functionality despite hardware limitations using C++ and **OOP**.
- Gained hands-on experience with **pneumatics** and its mechanical properties with programmed implementation.

FIRST Robotics Team 8729 Sparkling Youth Robotics Club

Ottawa, Ontario

Software Team Lead | Java, WPILib, PathPlanner

June 2023 – July 2025

- Instructed **19** software members in programming and testing of command-based robots using **OOP** with **Java**.
- Developed and tested autonomous paths with swerve using Bezier curve trajectories for precision in **PathPlanner**.
- Integrated computer **vision pipelines** using Raspberry Pi 4 and **PID tuning** using hardware systems like REV Hardware Client and FRC Driver Station.

Projects

Nodality.ai | React (Flow), Javascript, Python, PostgreSQL, Docker

Oct 2025 - Present

AI Mind-Mapping Full Stack Web Application

- AI focused mind-mapping application that makes suggestions separate user ideas and simulates graphs using **React** with React Flow node handling front-end and Node data handling with python implementation of OpenAI API.
- Developed a **Node.js/Express** backend with **PostgreSQL** to manage secure user authentication via session-based data.
- Implemented a **similarity index** to analyze semantic relationships between user idea nodes, allowing intelligent node clustering and AI-generated suggestions.

navigator.ai | Javascript, HTML/SCSS, Three.js, Leaflet.js, Python

Sept 2023 - July 2024

Traffic Routing FOSS Web Application

- Built a web-based navigation app optimizing routes by time, distance, and safety using **7 years** of Ottawa traffic data.
- Integrated **OpenOttawa API** for live traffic data and mapped incidents using **Three.js** and **Leaflet.js**.
- Designed an AI feedback system enabling users and city admins to improve road safety with **OpenAI API**.
- Featured in **2** major local news outlets **Ottawa Citizen** and **CTV primetime** in enhancing urban mobility innovation.